

# Sequentialization for Multiplicative Proof-Nets

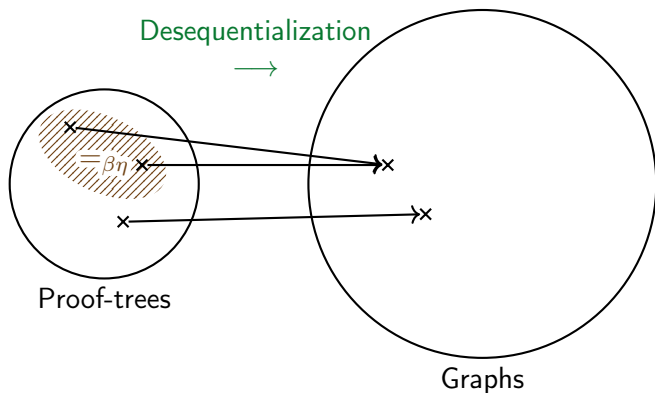
Rémi Di Guardia

1<sup>st</sup> October 2026



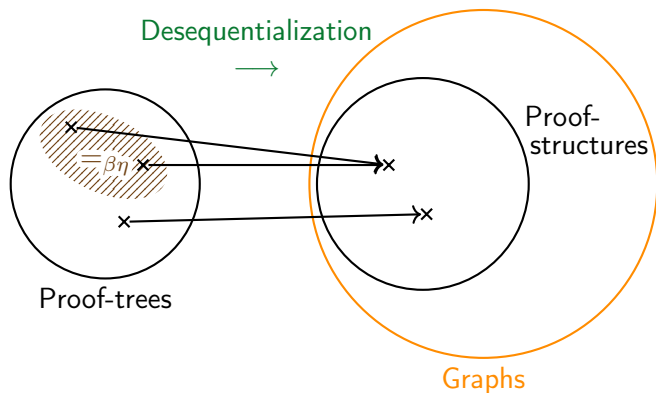
# From Proof-Nets back to Proof-Trees

Proof-Nets graphical syntax for proofs of **Linear Logic**, *canonical*



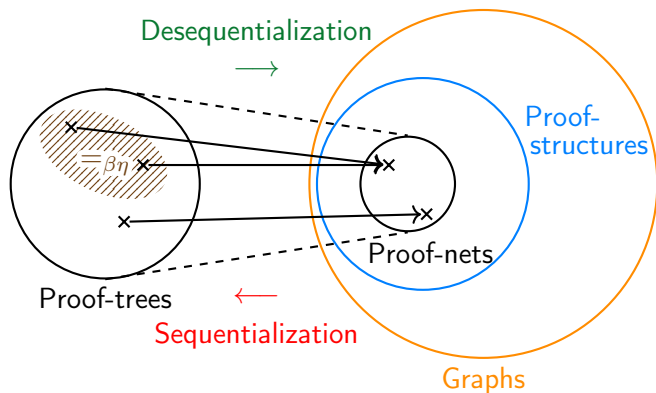
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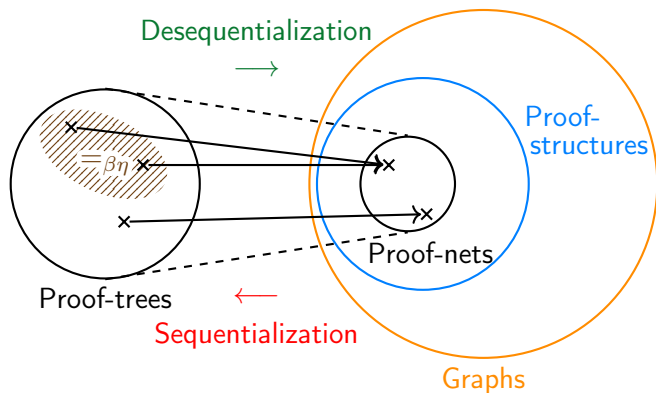
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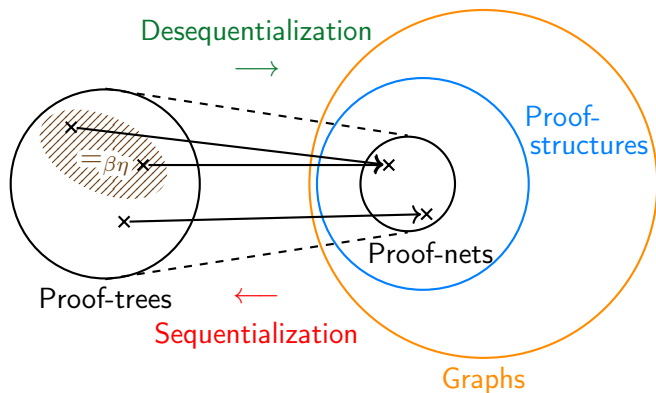
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Multiple **correctness criteria**, proofs of sequentialization

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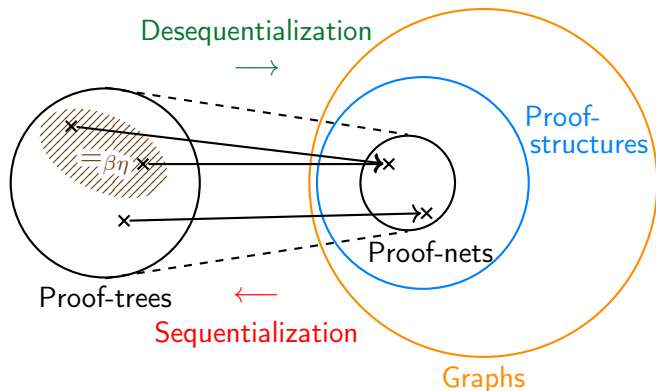


Multiple **correctness criteria**, proofs of sequentialization

Today an “easy” proof of sequentialization

# Outline

- ▶ Reminder: Deserialization & Proof-Nets
- ▶ Proof of Sequentialization



# Unit-free Multiplicative Linear Logic

## Formulas

$$A ::= X \mid X^\perp \mid A \otimes A \mid A \wp A$$

## Negation

$$(X^\perp)^\perp = X \quad (A \otimes B)^\perp = A^\perp \wp B^\perp \quad (A \wp B)^\perp = A^\perp \otimes B^\perp$$

## Sequents

$$\vdash A_1, A_2, \dots, A_n$$

## Rules

$$\frac{}{\vdash A^\perp, A} \text{ (ax)} \quad \frac{\vdash A^\perp, \Gamma \quad \vdash A, \Delta}{\vdash \Gamma, \Delta} \text{ (cut)}$$

$$\frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} \text{ } (\otimes) \quad \frac{\vdash A, B, \Gamma}{\vdash A \wp B, \Gamma} \text{ } (\wp)$$

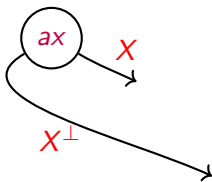
# Desequentialization: From Trees to Graphs

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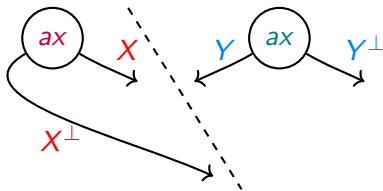
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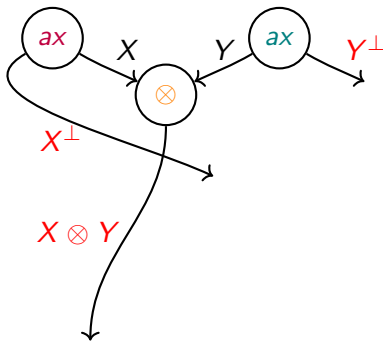
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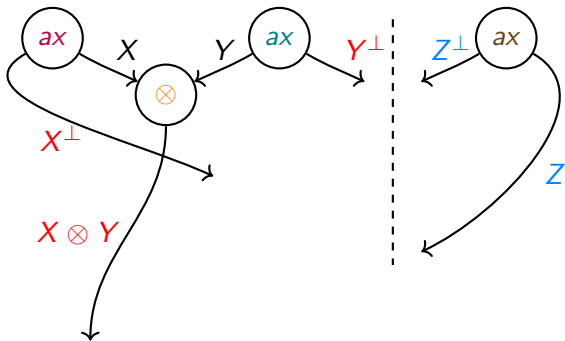
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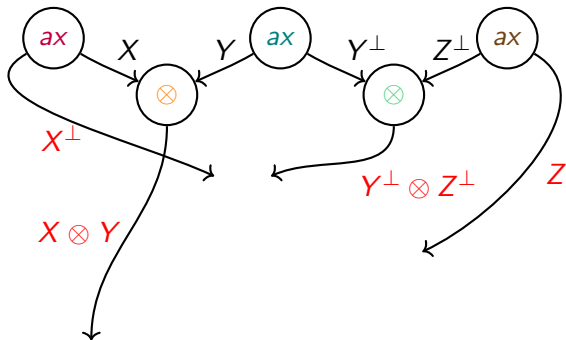
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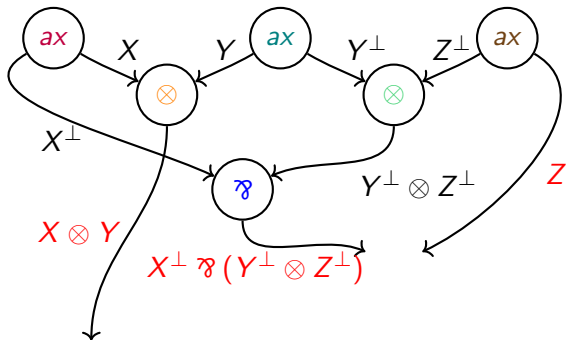
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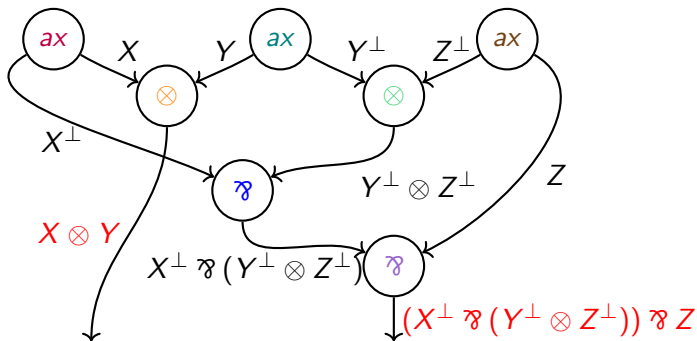
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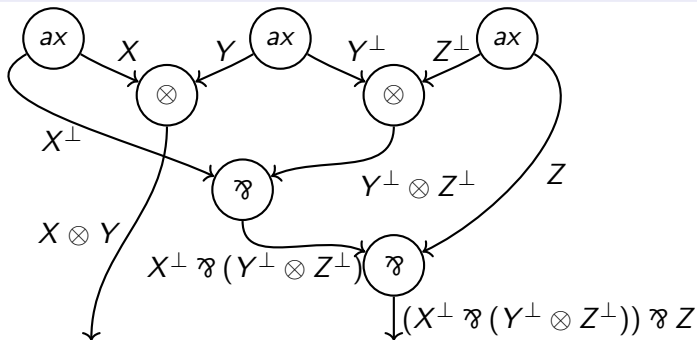
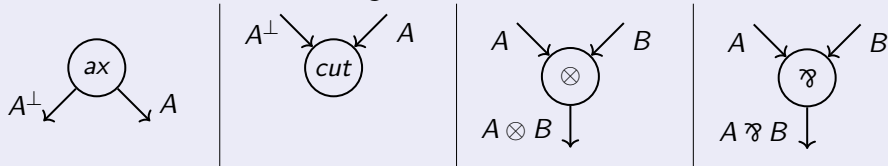
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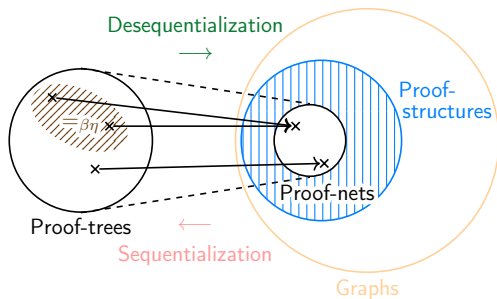
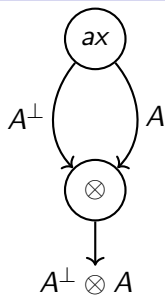
# Proof-Structure

## Definition: Proof-Structure

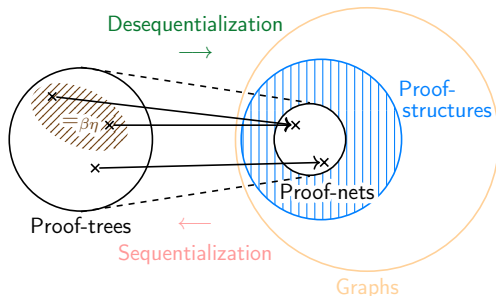
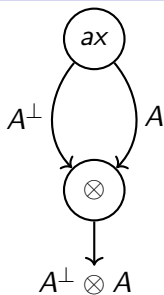
Partial graph with labels on vertices  $\rightarrow ax, cut, \otimes, \wp$   
 on edges  $\rightarrow$  formula



# Correctness Criterion

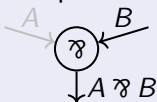


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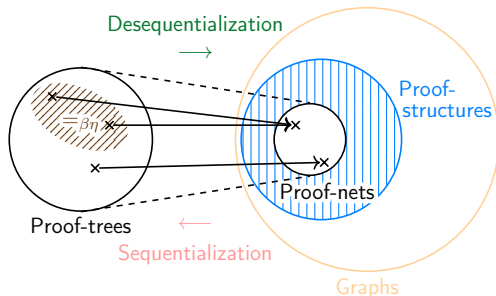
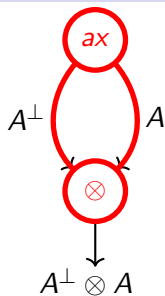
## Definition: Danos-Regnier Correctness Criterion

**Correctness graph:** remove one premise of each  $\wp$



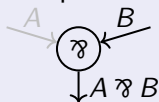
**Correct** when all correctness graphs are *acyclic* and *connected* (non-oriented paths)

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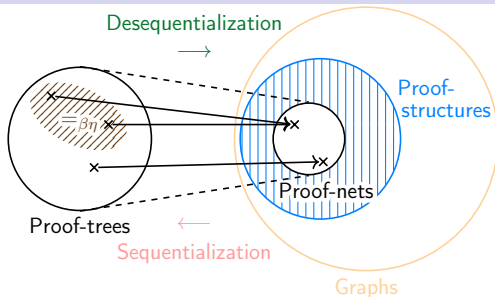
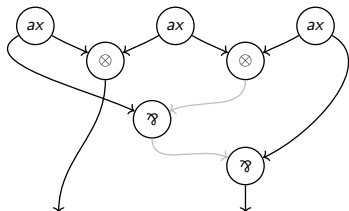
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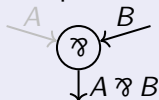


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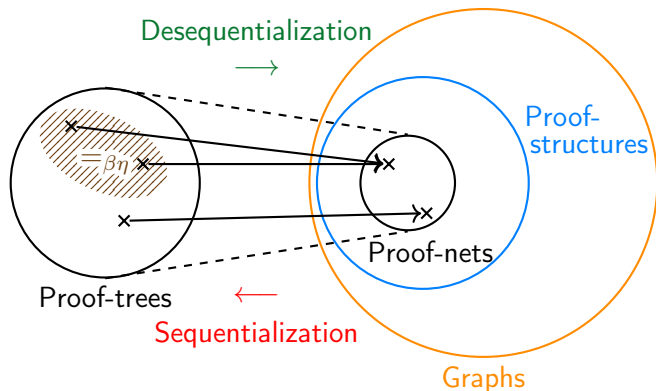


**Correct** when all correctness graphs are **acyclic** and **connected** (non-oriented paths)

**Switching path:** does not contain the two premises of a  $\wp$   
 correctness graphs acyclic  $\implies$  **No switching cycle**

# Outline

- ▶ Reminder: Deserialization & Proof-Nets
- ▶ Proof of Sequentialization



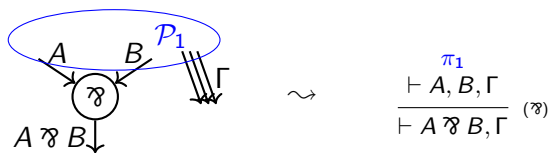
# Sequentialization

## Theorem: Sequentialization

Given a correct proof-structure, there is a proof-tree desequentializing to it.

How to prove it? Find a vertex corresponding to a **last rule**

- a  $\wp$  with no vertex below



- a  $\otimes$  with no vertex below and not in a cycle



- a *cut* not in a cycle  
 $\longrightarrow$  same as a  $\otimes$ , we will not consider *cut*

# Building a Switching Path

Idea of the proof build a *path*  $p$  by following a chain of **dependencies**  
 $\leadsto$  before doing this  $\otimes$ -rule we must first do this  $\wp$ -rule, before which we  
must do this  $\otimes$ -rule ... stopping on a rule we can actually do  
*(c.f. previous drawings)*

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## Plan

- 1 We build a **switching path**  $p$  by visiting vertices, halting our construction only on a vertex corresponding to a last rule.
- 2 We prove  $p$  **cannot “loop”**: its vertices are distinct.

Then we are done!

- The graph is finite  $\longrightarrow$  the path's construction must stop, necessarily on a vertex corresponding to a last rule!
- We remove this vertex and conclude by induction hypothesis.

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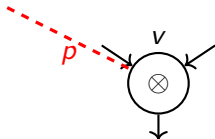
- The graph is finite  $\longrightarrow$  the path's construction must stop, necessarily on a vertex corresponding to a last rule!
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Initial  $p$  a premise of any arbitrary  $\otimes$  or  $\wp$  vertex  $v$  with no vertex below it  
(if only  $ax$  vertices, then easy to conclude)

## Ascending Path $\alpha$ : a $\otimes$ blocked by a $\wp$

Assume  $p$  ends with a premise of a vertex  $v$  with no one below it.

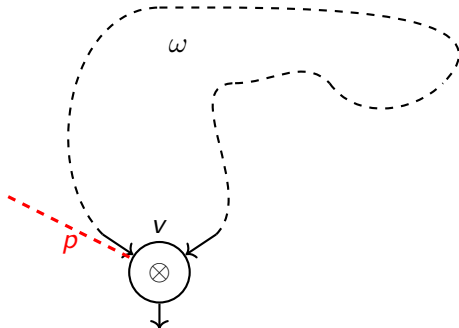
- If  $v$  is a  $\wp$  or is a  $\otimes$  not in a cycle: we are done.



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- Otherwise:  $v$  is a  $\otimes$  in a cycle  $\omega$ .

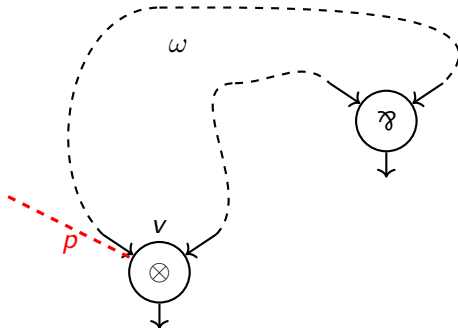


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Remember  $\omega$  cannot be switching: it has at least one  ;  
w.l.o.g.  $\omega$  has a minimal number of those (used later to not loop).

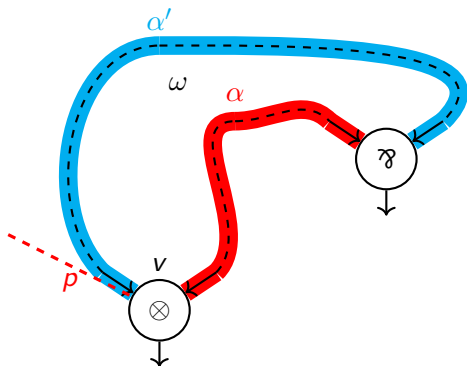


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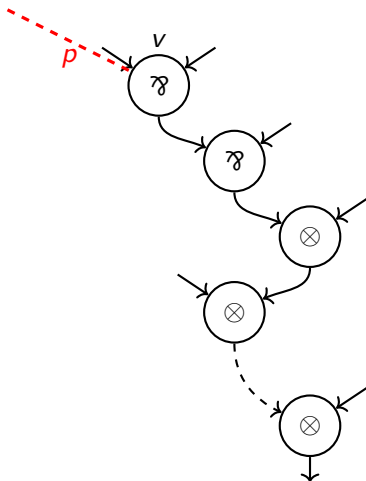
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- Otherwise:  $v$  is a  $\otimes$  in a cycle  $\omega$ .

Remember  $\omega$  cannot be switching: it has at least one  $\wp$  ;  
w.l.o.g.  $\omega$  has a minimal number of those (used later to not loop).  
Extend  $p$  with the sub-path  $\alpha$  of  $\omega$  towards the first  $\wp$  .



# Descending Path $\delta$ : formula dependency

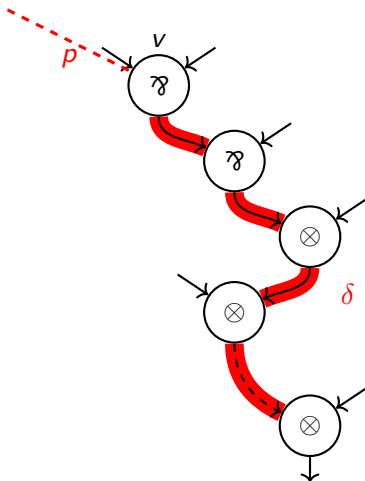
Assume  $p$  ends with a premise of a  $\mathcal{F}$ -vertex  $v$  with someone below it.



# Descending Path $\delta$ : formula dependency

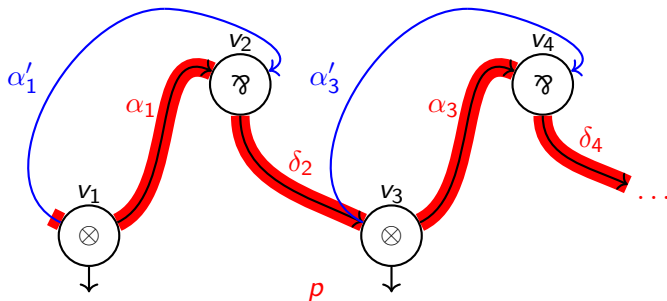
Assume  $p$  ends with a premise of a  $\mathcal{T}$ -vertex  $v$  with someone below it.

Extend  $p$  with **downward edges  $\delta$**  towards a vertex with no one below it.



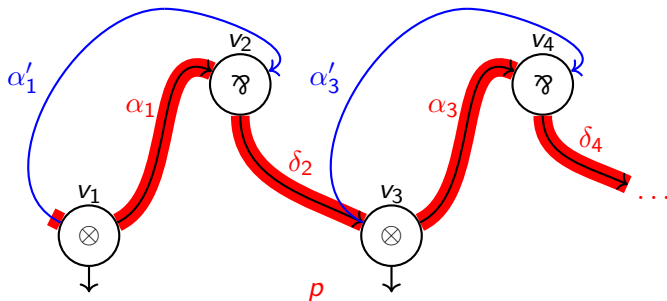
# Built Switching Path $p$

Concatenation of an ascending path  $\alpha_1$ , a descending path  $\delta_2$ , an ascending path  $\alpha_3$ , a descending path  $\delta_4$ , ...



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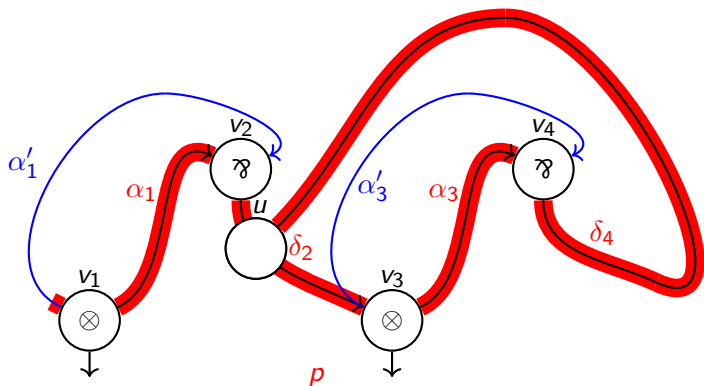
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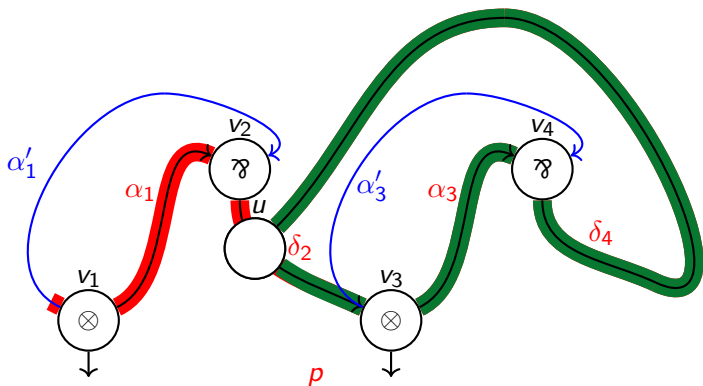
## Plan

- 1 We build a **switching path**  $p$  by visiting vertices, halting our construction only on a vertex corresponding to a last rule. ✓
  - 2 We prove  $p$  **cannot “loop”**: its vertices are distinct.
- We get a contradiction by looking at the first vertex  $u$  appearing twice.

# No Going Back on a Descending Path



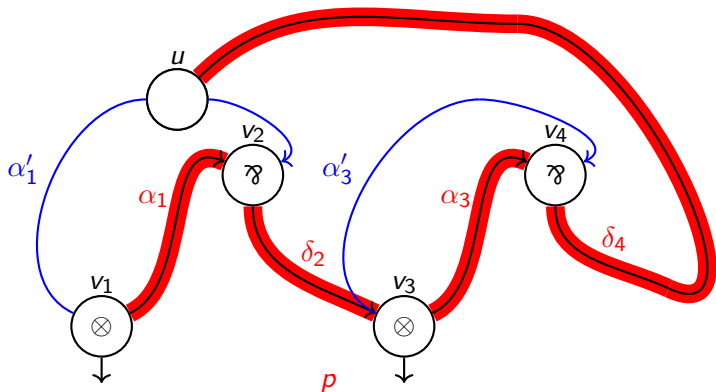
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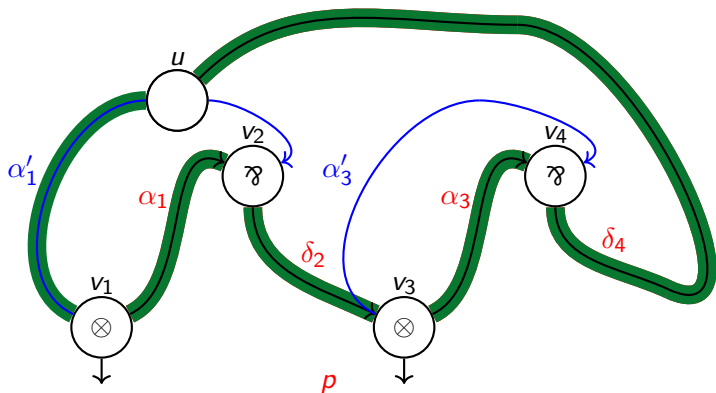
This is a switching cycle!

(no  at  $u$  as we took a downward edge in  $\delta_2$ )

# No Going Back on an Ascending Path('s cycle)



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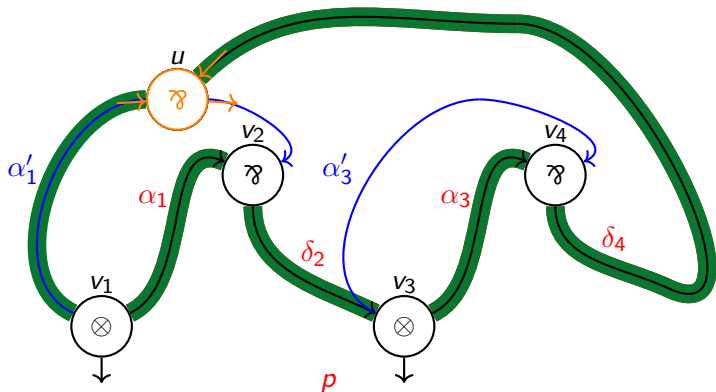


This cycle must have at least as many  as  $\alpha_1 \cdot \alpha'_1!$

Counting argument: since we lost one at  $v_2$ :

- there is no  between  $u$  and  $v_2$
- the green cycle takes both premises of  $u$  which is a  $\wp$

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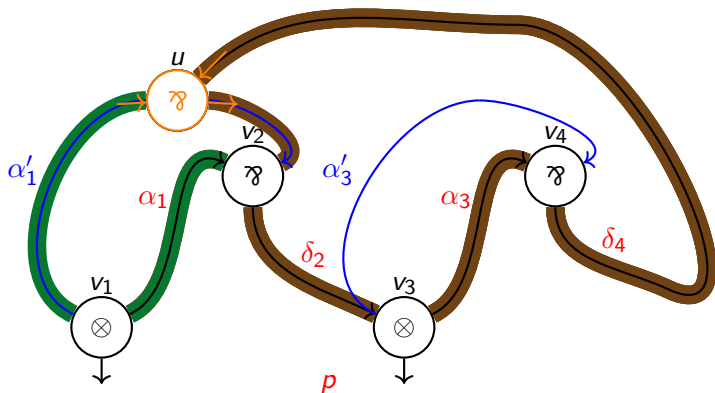


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Counting argument: since we lost one at  $v_2$ :

- there is no  between  $u$  and  $v_2$
- the green cycle takes both premises of  $u$  which is a  $\gamma$

This is a switching cycle!

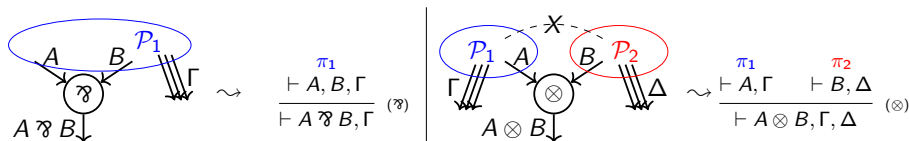
# Summing-up

## Plan

1 We build a **switching path**  $p$  by visiting vertices, halting our construction only on a vertex corresponding to a last rule. ✓

2 We prove  $p$  **cannot "loop"**: its vertices are distinct. ✓

→ we get a  $\otimes$  or  $\wp$  corresponding to a last rule, remove it and conclude by induction hypothesis



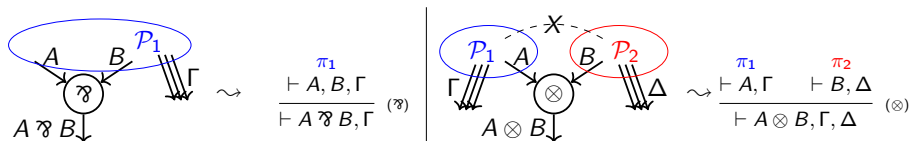
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## Plan

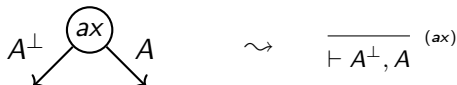
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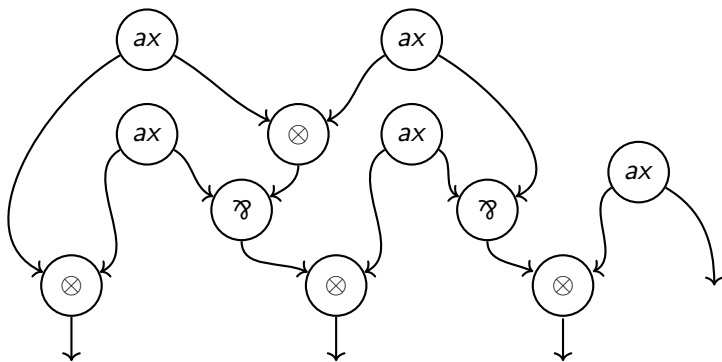
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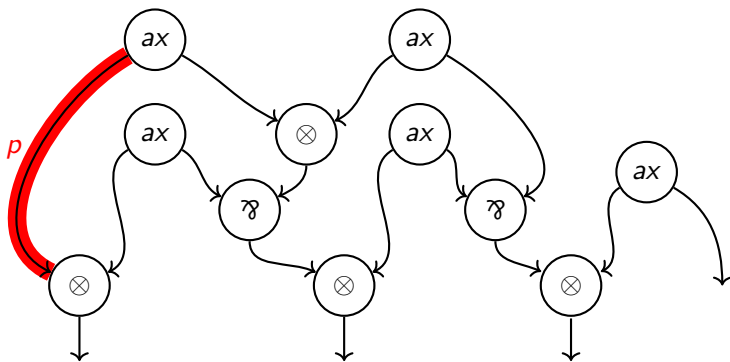
- This solves the case when the proof-net has a  $\wp$  or a  $\otimes$
- If it has only  $ax$ , easy: by connectivity it has exactly one, this is the image of an  $ax$ -rule



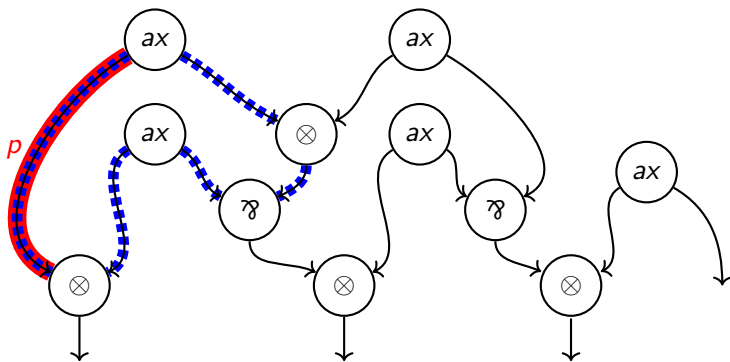
# Sequentialization's Path on an Example



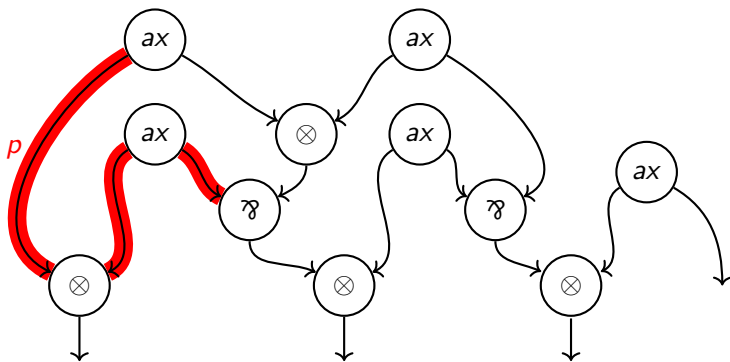
# Sequentialization's Path on an Example



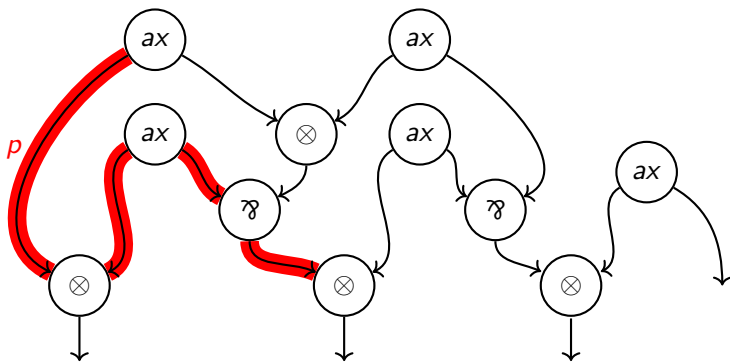
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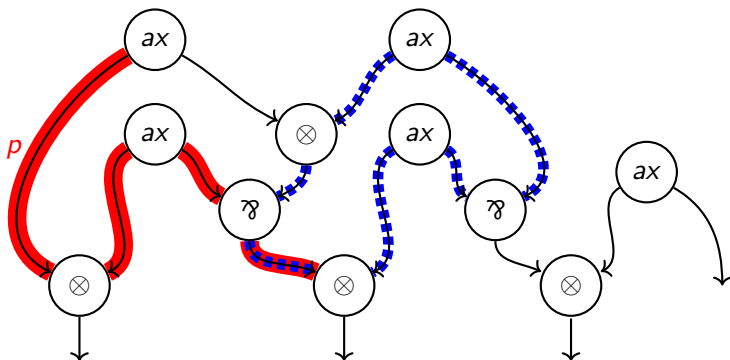
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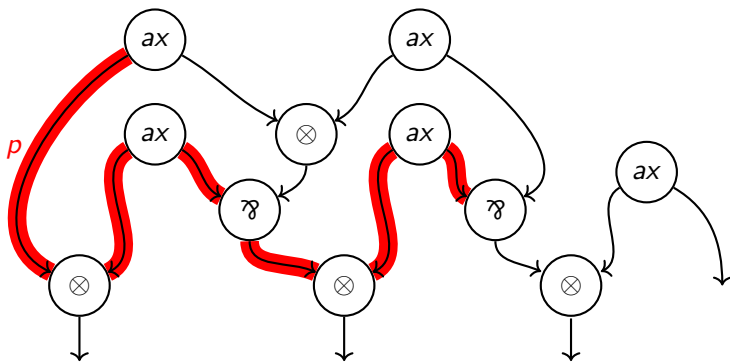
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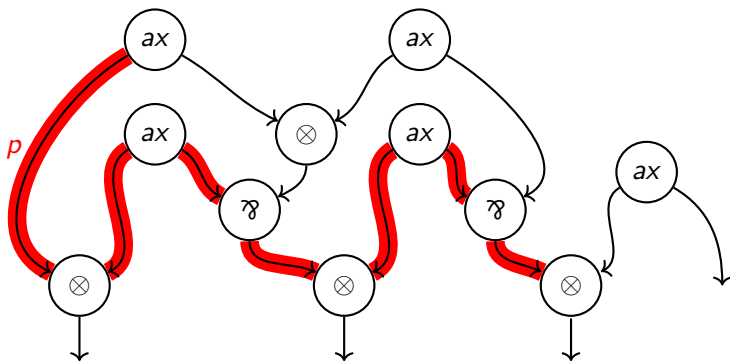
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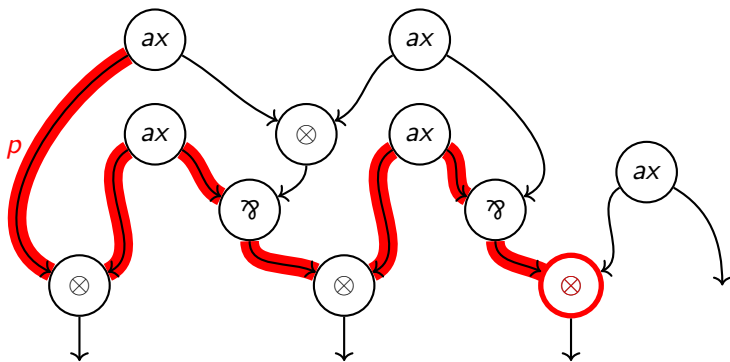
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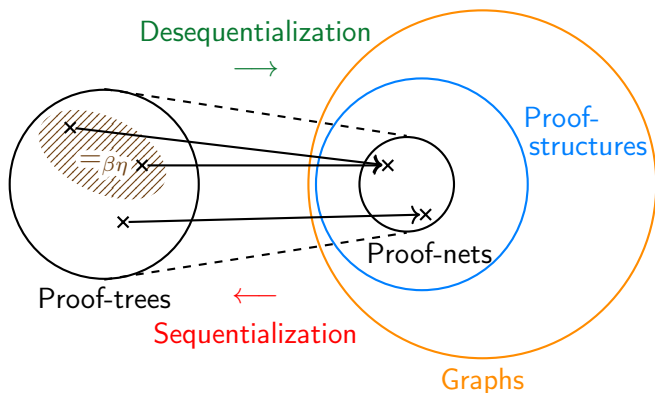


# Sequentialization's Path on an Example



# Concluding Remarks

- We only used acyclicity, not connectivity  
→ this proof extends to **MELL proof-nets** (with/without *mix*)!
- Proof formalized in  **ROCQ**
- Strong links with more standard **graph theory**: Kotzig's theorem on perfect matchings, Yeo's theorem on edge-colored graphs, ...
- The **quotient** of proof-trees by proof-nets is well-known



# References



Link to these slides:



 **Handbook of Linear Logic (draft)**, <https://ll-handbook.frama.io/ll-handbook/ll-handbook-public.pdf>

<https://ll-handbook.frama.io/ll-handbook/ll-handbook-public.pdf>

↪ [this proof written with a different presentation](#)

 **Yeo's Theorem for Locally Colored Graphs: the Path to Sequentialization in Linear Logic**, *Di Guardia, Laurent, Tortora de Falco, Vaux Auclair*, <https://doi.org/10.4230/LIPIcs.FSCD.2025.16>

↪ [this proof written in graph theory](#)

 **Yeo's Theorem for Locally Colored Graphs: the Path to Sequentialization in Linear Logic (in publication)**, *Di Guardia, Laurent, Tortora de Falco, Vaux Auclair*, <https://arxiv.org/abs/2603.03262>

↪ [extension to MALL proof-nets](#)